



Mathematical foundations in data science

Probability Review

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Course Overview



- Objective: The course will introduce students to the basic concepts of data analysis and pave the way for understanding more advanced topics in analytics and data science.
- An opportunity for a new start on statistics and probability
- An imperative lens to view all forms of decision-making in management. It is a way of thinking, even if it is not practiced in a prescriptive way.



Dovetailing with the previous knowledge



- You now have an understanding of working with Data through descriptive statistics. We will not revise that.
- You have been exposed to the basics of probability. We will partially revise, and add some new content.
- The big picture.



- The role of probabilistic modelling, descriptive and inferential statistics



Random Variables



- What are they? Revision
 - Discrete and continuous random variables
 - Moments
 - Joint distributions
 - Functions of Random Variables and simulation examples
- Specific distributions
 - Bernoulli, Uniform (discrete and continuous), Binomial, Poisson, Geometric, Exponential, Hypergeometric, Negative Binomial, Normal or Gaussian.



Basic definitions: Revision



- Random process or experiment: A process/phenomenon/experiment which can result in different outcomes, even though it is repeated in the same manner every time.
- Sample spaces: The set of all possible outcomes of a random experiment is called a sample space.
- Toy examples: Tossing a coin or dropping a ball from a window.



Basic definitions: Revision



- Random variables

- Moving from tables to expressions: Let us toss a coin 5 times

Outcomes (# of heads)	Probability
0	3%
1	16%
2	31%
3	31%
4	16%
5	3%



$$f(x; n, p) = P(X = x) = {}^n C_x \cdot p^x (1 - p)^{n-x}$$

- Random variable: the uncertain outcome of a random process, often denoted by capital letters
- What is small x ?
- The Probability Mass Function (PMF) is the expression itself ($f(x; n, p)$)



Basic Definitions: Revision



- Continuous and Discrete random variables
 - Take our examples of tossing coins and dropping balls. What is different?
 - Is it about integers vs decimals? Is it about infinite and non-infinite values? Countable vs non-countable.
 - In the discrete case, the PMF reflects the probability that a specific value or a countable set of values can occur.
 - In the continuous case, the Probability Density Function (PDF) reflects probability of occurrence within a given range or interval.
 - Absolute versus relative likelihood: A random variable is characterized by a mathematical function (PDF/PMF) that quantifies the likelihood (absolute or relative) of the outcomes in the sample space of a random experiment.
 - All RVs can be characterized by a Cumulative Distribution Function (CDF) which captures the probability of occurrences below a certain value x . $F(x) = P(X \leq x)$.



Basic Definitions: Examples



- Having too many discrete states, and the discretization of a continuous random variable.
- Examples:

Area	Discrete	Continuous
Operations	Defective/Non Defective, Defects, Arrival in time.	Time to failure, performance/characteristic of a part (strength, stress, etc.), Demand
Finance	Letter based credit rating, delinquency/default, fraudulent transactions, M&A outcome	ROI as percentage, PnL, valuation
HR	Attrition/onboarding (small samples), team size, team success.	Attrition/onboarding (large samples), Employee salary, time to join, time to leave.
Marketing	Number of units sold, product likeability.	Time spent in a shop, Number of clicks/visits per site, lift scores.
Miscellaneous	Grades in this course, Rain or not, Covid test (+ve or -ve), outcomes of test match in cricket, which team will win a basketball tournament.	Marks in this course, Chemical composition, mileage of a vehicle,, rainfall in mm, # of Covid cases, Body temperature.



Simple ways of describing Random Variables

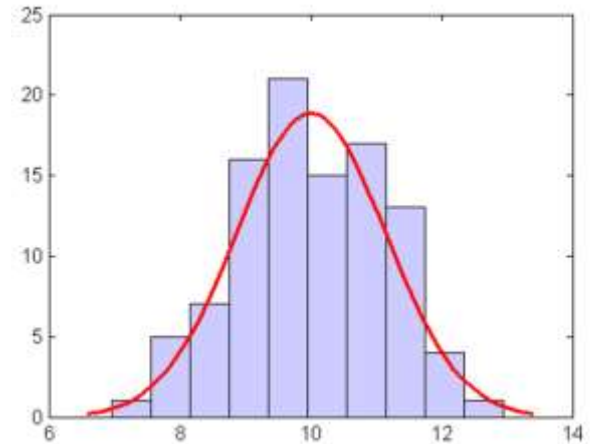


- We looked at a mathematical expression in the form of PMF and PDF
- Some basics:
 - For PMFs $f(x_i) = P(X = x_i)$ and a) $0 \leq f(x_i) \leq 1$, b) $\sum_{i=1}^n f(x_i) = 1$
 - Example: $f(x) = \frac{2x+1}{25}, x = 0,1,2,3,4$
 - For PDFs $P(a \leq x \leq b) = \int_a^b f(x) \cdot dx$ and a) $f(x) \geq 0$, b) $\int_{-\infty}^{\infty} f(x) = 1$, c)
The probability at a given point is 0
 - Other descriptions are CDF, and the four moments: Mean, Variance, Skew and Kurtosis

Four Moments

- Remember the histogram from pre-term?
- Remember Mean and standard deviation from pre-term?
 - You calculated that from data (empirical approach)
 - What if I gave you a PDF/PMF (theoretical)
- Measures of Central Tendency (Mean)
- Dispersion (Variance)
- Skew and Kurtosis

Data Set
10.04
9.31
11.15
11.22
10.19
10.49
8.38
10.32
8.14
7.89
10.07
10.42
11.55
9.63
9.05
8.96
12.57
.
.
.





Revision: Measures of Central Tendency (from data)



- Data Set: 3,4,3,1,2,3,9,5,6,7,4,8
- Mean

$$\frac{3+4+3+1+2+3+9+5+6+7+8+4}{12} = 4.583 \quad \frac{x_1+x_2+x_3+\dots}{n} \quad \text{or} \quad \frac{\sum_{i=1}^n x_i}{n}$$

- Median

1,2,3,3,3,4,4,5,6,7,8,9 Hence Answer = 4

- Mode

The value 3 appears 3 times, and 4 appears 2 times and all other values appear once. Hence 3 is the mode



Measures of Central Tendency

- Where do we want to use Mean, Median and Mode
- Choosing between mean and median
 - Bad outliers
 - Errors
 - Do not provide a realistic picture of the story
 - Good outliers
 - The story is in the outliers
- Mode
 - Useful with nominal variables
 - Multi modal distributions



Revision: Measures of Dispersion



- Data set: 3,4,3,1,2,3,9,5,6,7,4,8
- Range (Max-Min) ($9 - 1 = 8$)
- Inter Quartile Range: 3rd quartile - 1st quartile (75th Percentile – 25th Percentile) ($6.5 - 3 = 3.5$)
- Sample Standard deviation

$$\sqrt{\frac{1}{N - 1} \sum_{i=1}^N (x_i - \bar{x})^2}$$
$$= \frac{1}{12 - 1} \sum ((3 - 4.58)^2 + (4 - 4.58)^2 \dots)$$



Revision: Measures of Dispersion



- Questions that go with Standard deviation
 - Why do we use the square function on the deviations? What are its implications?
 - Why do we work on standard deviation and not the variance?
 - Why do we average by dividing by $N-1$ and not N ?
- Mean absolute Deviation and its variants
 - Use $|x_i - \bar{x}|$ instead of $(x_i - \bar{x})^2$



Overview of Mean and Standard deviation



	Empirical	Theoretical/Conceptual	
		Discrete	Continuous
Mean	$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$	$E(X) = \mu = \sum_{i=a}^b (x_i \cdot p_i)$	$E(x) = \mu = \int_a^b x f(x) dx$
Standard Deviation/ Variance	$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^n (x_i - \bar{x})^2}$ Or $s = \sqrt{\frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2}$	$\text{Var}(x) = \sigma^2 = \sum_{i=a}^b p_i (x_i - \mu)^2$ or $\sum_{i=a}^b p_i (x_i)^2 - \mu^2$	$\text{Var}(x) = \sigma^2 = \int_a^b (x - \mu)^2 f(x) \cdot dx$ or $\int_a^b (x)^2 f(x) \cdot dx - \mu^2$

- Theoretical standard deviation in terms of mean:

$$\begin{aligned} E((X - \mu)^2) = \\ E(X^2) - (E(X))^2 \end{aligned}$$



Example: COMPUTER SHIPMENTS & QUALITY

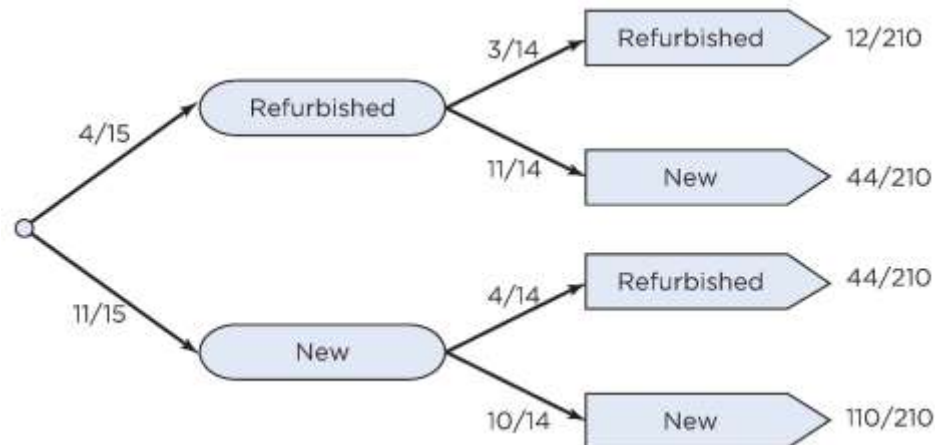


- Motivation

CheapO Computers shipped two servers to its biggest client. Four refurbished computers were mistakenly restocked along with 11 new systems. If the client receives two new servers, the profit for the company is \$10,000; if the client receives one new server, the profit is \$4,500. If the client receives two refurbished systems, the company loses \$1000. What are the expected value and standard deviation of CheapO's profits?

Example: COMPUTER SHIPMENTS & QUALITY

- Mechanics – Tree Diagram





Example: COMPUTER SHIPMENTS & QUALITY



- Mechanics – Compute μ and σ
- $E(X) = \mu = \$7,067$
- $\text{Var}(X) = \sigma^2 = 10,986,032 \2
- $\text{SD}(X) = \sigma = \$3,315$
- What if our example did not have a table?



General Operations on single Random Variables



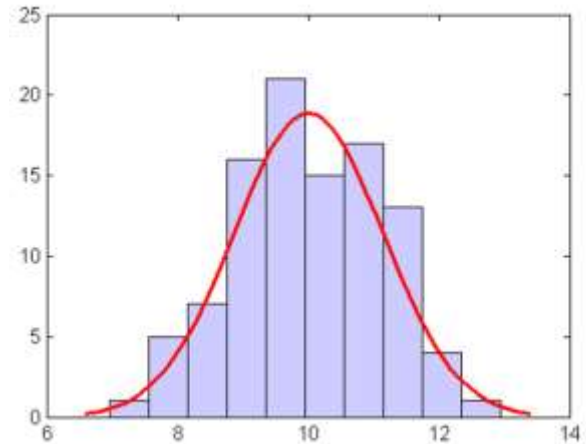
- Effect of adding or removing a constant
 - Mean $E(X \pm c) = E(X) \pm c$
 - Standard Deviation $Var(X \pm c) = Var(X)$; $SD(X \pm c) = SD(X)$
- Effect of multiplying or removing a constant
 - Mean $E(cX) = c E(X)$
 - Standard Deviation $Var(cX) = c^2 Var(X)$; $SD(cX) = |c| SD(X)$
- Putting these together
 - $E(a + bX) = a + bE(X)$
 - $SD(a + b X) = |b|SD(X)$
 - $Var(a + bX) = b^2Var(X)$

Skew and Kurtosis

- What about skew and kurtosis?
- Skew:
 - Left or right-tailed property
 - It reflects the asymmetric deviation from the mean
- Kurtosis
 - Fat tailed
 - If I make the tails fatter, am I not just increasing variance?

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.

Histogram





Joint Distributions



- Why do we need to look at distributions jointly.
 - Photocopy shop example
- Modified example from the text

		X (Revenue)			$p(y)$
		$x = 3$	$x = 5$	$x = 7$	
Y (cost)	$y = 2$	0.08	0.07	0.00	0.13
	$y = 4$	0.01	0.62	0.02	0.65
	$y = 6$	0.00	0.11	0.09	0.20
	$p(x)$	0.09	0.80	0.11	1

- Joint probability $P(X=3, Y=2)$
- Marginal probability $P(X=3)$
- Conditional probability $P(Y=2|X=3)$



Going from tables to formulas



- Joint

PDF $f_{X,Y}(x, y)$ captures $P(a \leq X \leq b, c \leq$



Properties of combining two random variables



- $P(X=3, Y=2)$ can be written $P_{i,j} = P_{1,1}$
- $P(X=3, Y=2)$ is the same as $P(X=3 \cap Y=2)$. If $X=3$ is an event (A) and $Y=2$ is an event (B) then $= P(A \cap B)$.
- From what we understood in conditionals it is clear that

$$P(A|B) = P(A \cap B)/P(B)$$

- Multiplication rule: $P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$
- Bayes theorem: $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$
- Two events are independent if $P(B|A) = P(B)$ or $P(A|B) = P(A)$ therefore $P(A \cap B) = P(A) \cdot P(B)$



Example problems using Bayes theorem



- We sell computers with optional insurance against damage. 90% of our customers do not take insurance. We find that those who don't purchase insurance are more careful with the product than those who do. There is a 30% chance that a customer with insurance will file a repair request, whereas only 5% of those without insurance will file a request.
- If we receive a repair request, what is the probability that it is from a customer who has insurance?



Examples using Bayes theorem



- Let us say a stock will go up if a merger goes through (down if it does not). Your insightful market research shows that there is an 80% chance of the merger occurring. Historically your research team has been good. However, you do not want to take a bet on just this information. You hire an insider to overhear the meeting involving the merger discussions. However, this informant's listening is only 90% accurate (or she sometimes mishears the discussion). Following the meeting, she calls you and tells you that the merger is not happening. What are the odds that she is correct?



Decision-making using EVPI and EVSI

Expected Value of Perfect Information (EVPI)

- EVPI is the expected price that we are willing to pay to gain access to perfect information.
- $EVPI = \text{Expected value with perfect information} - \text{Expected value without perfect information}$

Expected value of sample information (EVSI)

- EVSI is the expected increase in monetary value by obtaining access to sample of additional observations.
- $EVSI = \text{Expected value with sample information} - \text{Expected value without sample information}$

Example 1

- Consider a situation in which there is 80% chance that stock price goes up, and 20% chance that it goes down. The profit we make by investing when price is up is Rs.10, and the loss we incur by investing when price is down is Rs.20. Calculate EVPI
- The payoff matrix can be represented as follows:

Decision	Stock up	Stock down
Buy	10	-20
Not buy	0	0
Probability	0.8	0.2



Example 1



- Expected value without perfect information:
 - Expected monetary value = $(10 \times 0.8) + (-20 \times 0.2)$
(EMV) = 4
- Expected value with perfect information (EV|PI):
 - If it is known that the stock is up, we make a decision to buy. If it is down, we make a decision to not buy.
 - $EV|PI = (0.8 \times 10) + (0.2 \times 0) = 8$
- $EVPI = EV|PI - EMV$
 $= 8 - 4 = 4$



Example 1: EVSI scenario 1



- Considering Example 1, assume we have the stock price status for previous 5 years. From this sample of 5 years, 90% of time we predict the current status correctly. 10% of the time the prediction is wrong.
- That is, if the stock is actually up, we predict that it is up with probability 0.9, and predict that it is down with probability 0.1

Example 1: EVSI scenario 1

Decision	Stock up	Stock down
Buy	10	-20
Not buy	0	0
Probability	0.8	0.2

	Actual status	
	Up	Down
Sample info	Up	0.9 0.1
	Down	0.1 0.9

- $EV|SI = (0.8 \cdot 0.9 \cdot 10) + (0.1 \cdot 0.2 \cdot (-20)) = 6.8$
- We know that, $EMV = 4$
- $EVSI = EV|SI - EMV$
 $= 6.8 - 4 = 2.8$
- Note: EVPI value was 4

Example 1: EVSI scenario 2

- Suppose for example 1, the probabilities from the sample is modified by the given matrix.

		Actual status	
		Up	Down
Sample info	Up	0.7	0.2
	Down	0.3	0.8

- $EV|SI = (0.8 \cdot 0.7 \cdot 10) + (0.2 \cdot 0.2 \cdot (-20)) = 4.8$
- We know that, $EMV = 4$
- $EVSI = EV|SI - EMV$
 $= 4.8 - 4 = 0.8$

Decision	Stock up	Stock down
Buy	10	-20
Not buy	0	0
Probability	0.8	0.2

- A note on functions of random variables